

Why Avoid Collisions? Exploit Them! Information Collection in Multi-Tagged RFID Systems

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Abstract—We investigate the problem of target object information collection in multi-tagged RFID systems. Different from its single-tagged peers, the multi-tagged RFID scenario introduces three new challenges: 1) Tags on the same object carry the same information, so reusing single-tagged algorithms causes unnecessary redundancy; 2) The transition of objects from being tagged one to multiple tags leads to an upsurge in slot collisions; 3) Gathering information from more tags necessitates more downlink transmission, making it hard to limit broadcast information while ensuring time-efficient information collection. To tackle these technical challenges, we propose an efficient information collection algorithm, called Backtracking Collision Peeling (BCP), featuring three key techniques. First, BCP selects a single time slot (*allowing even collision slots*) for each target object to convey its information. This approach bypasses the need for a high-latency collision elimination process, thereby significantly reducing time overhead. Second, by exploiting dependencies among the selected slots, BCP recovers object information in complex collision slots using object information from already resolved slots, offering a novel and effective solution for handling signal collisions. Third, to minimize downlink transmission cost, BCP polls tags by transmitting only incremental changes between polling vectors, rather than the complete vectors. We further improve performance with E-BCP by enhancing the utilization of collision slots, thereby reducing the number of required polling rounds. Experiments show BCP and E-BCP outperform existing methods by at least 35% in aggregate execution time, while also exhibiting stronger stability and robustness.

Index Terms—Information collection, multi-tagged RFID systems, throughput optimization.

I. INTRODUCTION

CHARACTERIZED by its non-line-of-sight tracking capability [1], [2], low-deployment cost [3], and battery-free operation [4], radio frequency identification (RFID) technology has been regarded as a promising enabler for a broad spectrum

of Internet of Things (IoT) applications [5], [6], ranging from inventory control and warehouse management [7] to behavior recognition and object localization and tracking [8]. A standard RFID system consists of a reader and a large number of passive tags. These tags harvest energy from the RF signals emitted by the reader and communicate with it by backscattering the signals [9], enabling efficient and scalable identification without onboard power sources [10], [11].

In large-scale RFID systems, object monitoring is often hindered by challenging environmental factors such as extreme temperatures and exposure to corrosive substances like strong acids or bases [12], [13]. Statistically, according to a practical report by IDTechEx [14], Walmart's RFID applications achieve object detection rates of 90% at the case level and 95% on conveyor belts. The rate, however, drops significantly to only 66% for fully loaded pallets, primarily due to adverse environmental conditions. To upgrade system reliability and detection accuracy, multi-tagged RFID systems have emerged as a promising solution, where each object is attached with multiple tags. Even if a subset of tags on an object fails, the remaining ones can continue to monitor this object, thereby reducing the risk of data loss or tracking failure.

Despite its paramount importance in improving monitoring accuracy and system robustness, the multi-tagged RFID system remains largely underexplored in the current literature. Only a limited number of studies have addressed the problem of missing tag identification in such scenarios [13], [15], while other critical research issues remain largely unexamined.

In this paper, we investigate a practically important problem in multi-tagged RFID systems: how to collect information from a given subset of multi-tagged objects in the most time-efficient manner. This problem arises in a wide range of RFID-enabled applications [9], [16], [17], [18]. For instance, in supermarkets or warehouses where each product is attached with multiple tags, managers need to retrieve the expiration dates of a specific subset of products to update inventory records efficiently [19]. Different from its single-tagged peers, the multi-tagged RFID scenario introduces three new challenges in designing an object information collection algorithm:

- *Tag redundancy*: Since tags attached to the same object carry identical information, a response from just one tag per target object is sufficient for information collection. Hence, it is crucial to carefully select one tag from each target object to maximize time efficiency.
- *Collision surge*: As objects transition from being tagged with a single tag to multiple ones, the increased number

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of tags leads to more slot collisions, rendering information collection even more challenging.

- *System throughput*: Collecting information from more tags necessitates increased downlink transmission. Therefore, we should limit the amount of broadcast information while ensuring successful information collection.

Driven by the above technical challenges, in this paper we study information collection in multi-tagged RFID systems by making three contributions:

- *Efficient tag redundancy elimination*: We propose an efficient information collection algorithm, Backtracking Collision Peeling (BCP). Unlike conventional methods that select one tag from each target object to report information, BCP selects a single time slot for each target object. As the selected slot can be a collision slot, BCP circumvents the need for a high-latency collision elimination process, thereby significantly reducing overall time overhead.
- *Fully utilization of collision slots*: By exploiting the inherent dependencies among selected slots—including collision slots—BCP leverages object information obtained from already resolved slots to extract object information in more complex collision slots. This approach provides a novel and effective solution for handling signal collisions.
- *Significant improvement in system throughput*: To reduce downlink transmission overhead, BCP polls tags in the selected slots by broadcasting only the differential information between consecutive polling vectors, rather than the complete vectors. To further enhance performance, we introduce an enhanced version, E-BCP, which expands the effective use of collision slots and hence reduces the number of required polling rounds. Experimental results demonstrate that both BCP and E-BCP outperform the best existing solutions by at least 35% in aggregate execution time, while also exhibiting greater stability and robustness.

II. BACKGROUND AND PROBLEM FORMULATION

A. System Model

We consider a time-synchronized multi-tagged RFID system consisting of a reader, a set $\mathcal{N}$ of N tags, and a set $\mathcal{E}$ of E physical objects. The reader is connected to a back-end server with powerful computing and storage capabilities. For simplicity, we treat the server and the reader as an entity and just call it the reader. Each tag in $\mathcal{N}$ carries a unique ID, and each object in $\mathcal{E}$ is attached by n distinct tags. Besides, each tag stores l -bit information about the object it is attached to (e.g., the manufacturer of objects) in its user memory.

The reader communicates with tags via the standard Framed Slotted Aloha (FSA) protocol, which strictly follows the rule of “listen-before-talk”. Specifically, the reader first broadcasts commands and parameters including a frame size f and a seed s to tags. After that, each tag exploits its ID and the received seed s to generate a pseudo-random value via a uniform hashing function $H[(ID, s) \bmod f]$. In the FSA protocol, time slots are categorized into *empty*, *singleton*, and *collision* slots: In empty slots, no tag responds; in singleton slots, only one tag responds;

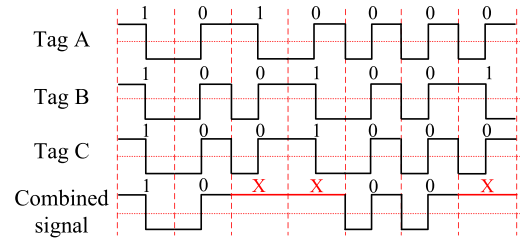


Fig. 1. Illustration of Manchester coding.

a slot is called a g -collision slot if g ($g \geq 2$) tags respond in this slot.

We make two reasonable assumptions in our system model. The first is that tags attached to the same object carry identical l -bit object information. This is justified because the information of an object (e.g., date of manufacture, category, or brand) is shared among the tags by which it is attached. The second is that all tags' IDs are apriori known to the reader, which is a fundamental assumption in the RFID information collection problems and has been broadly applied in existing RFID research [9], [10], [16], [17], [18], [20], [21], [22]. This assumption is practical: in real-world RFID applications, such as logistics and warehouse management, tag IDs are typically collected during object check-in.

B. Signal Encoding

At the physical layer, we adopt standard Manchester coding, which is also widely applied in contemporary RFID systems. The main motivation for using Manchester coding is its ability to decode a specific signal from a composite signal when other signals in the composite signal are available. As depicted in Fig. 1, in Manchester coding, “1” and “0” are represented by high-to-low and low-to-high signal transitions, respectively. Tags A, B, and C store three pieces of 7-bit object information “1010000”, “1001001”, and “1001000”, respectively. If they select the same slot to transmit their stored object information utilizing Manchester coding, the combined signal received by the reader is “10XX00X”, where “X” denotes a collided bit. As collided bits in the combined signal occur exclusively at positions where the three tags' responses differ, the generation of a collided bit is equivalent to a bitwise XOR operation among the three tags' responses. Hence, if the reader has prior knowledge of any two tags' responses, it can extract the third tag's response from the combined signal based on the rationale of Manchester coding.

A critical requirement for the above process is to guarantee the synchronous backscattering of signals from tags within the same time slot. [23] experimentally demonstrates that the synchronization offset in commercial off-the-shelf (COTS) tags typically remains within $1 \mu\text{s}$, meaning only 5.3% of a single bit's transmission duration. Furthermore, the feasibility of bit-level synchronization in RFID systems has been empirically validated through real-world experiments using USRP and WISP tags [24], [25].

C. Problem Formulation

This paper studies the target multi-tagged object information collection problem in RFID systems, which is formally defined as: *Given a set $\mathcal{T}$ of T target objects ($\mathcal{T} \subset \mathcal{E}$), we seek to devise an RFID information collection algorithm to minimize the expected execution time in gathering the l -bit information from all T target objects in $\mathcal{T}$.* A typical scenario corresponding to our problem is the RFID system in supermarkets, where each good is attached with multiple tags and the manager needs to gather the expiration date of a subset of the goods to update their inventory statistics.

This problem appears similar to the prior tag category/group information collection problem, where tags within the same group can be considered attached to the same object. However, in our scenario, tags on the same object do not share identical category IDs. As a result, existing category-level information collection algorithms [20], [26], which rely on category IDs, are inapplicable to our problem. This underscores the necessity for a novel algorithmic approach specifically tailored to address the constraints of our problem.

III. TARGET OBJECT INFORMATION COLLECTION BASED ON BACKTRACKING COLLISION PEELING (BCP)

In this section, we design a novel target object information collection algorithm for multi-tagged RFID systems, termed backtracking collision peeling (BCP), outperforming all existing RFID monitoring solutions applicable to our problem. The pivotal improvement over existing multi-tagged RFID monitoring algorithms [13], [15], [23] is three-fold.

- *Utilization of collision slots:* Unlike major existing multi-tagged RFID monitoring algorithms that rely solely on singleton slots and avoid collision slots, our algorithm BCP fully leverages the two categories of slots to enhance the throughput within a frame.
- *Efficient information extraction:* By exploiting the inherent dependencies among distinct types of slots—including collision slots—BCP leverages object information obtained from already resolved slots to extract object information in more complex slots. This approach provides a novel and effective solution for handling signal collisions.
- *Low-latency incremental polling:* BCP polls tags in the selected slots by broadcasting only the differential information between consecutive polling vectors rather than transmitting the complete vectors, significantly optimizing system throughput.

A. Design Rationale

In multi-tagged RFID systems, to gather object-level information from a particular object, a response from just one tag attached to that object is adequate. This tag is referred to as a *representative tag*. Existing multi-tagged RFID algorithms [17], [18], [23] typically select a tag hashed to a singleton slot from an object to serve as the object's representative tag, while keeping other tags on the object silent. Consider Fig. 2 as an example

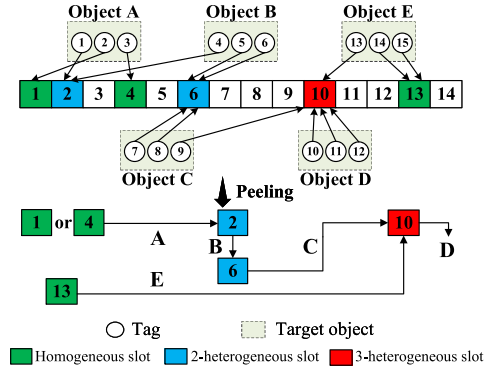


Fig. 2. Design rationale behind BCP algorithm.

where 15 tags attached to 5 target objects are hashed to a frame. Existing algorithms generally select the tag 2 or 3 as the representative tag of object A and the tag 14 or 15 as the representative tag of object E to convey information. However, these algorithms fail to identify a representative tag for the other three target objects, as all tags on these objects are hashed to collision slots. This reduces throughput in the frame and hence increases the time overhead. While multi-round hashing might alleviate this issue by increasing the proportion of singleton slots in a frame, it incurs substantial additional computation and communication overheads, especially in large-scale RFID systems.

Motivated by these limitations, we propose selecting a slot, termed a *representative slot*, rather than a tag, from each target object to convey its information. Specifically, in Fig. 2, we select slots 1 (or 4), 2, 6, 10, and 13 as representative slots to gather information from the five target objects. Since slot 1 is a singleton slot, the reader can easily collect object A's information from this slot. Additionally, by utilizing the capability of Manchester coding to decode a signal from a combined signal when already accessing other signals in the combined signal, the reader can extract object B's information by peeling object A's information from the combined signal in slot 2. We refer to this procedure as *collision peeling*. Similarly, the information of objects C and D can be extracted by performing collision peeling on slots 6 and 10, respectively, as indicated by the arrows in Fig. 2. Through this method, BCP significantly enhances the throughput within a frame and reduces the number of frames required, as well as the aggregate overhead of initializing and terminating frames.

B. Algorithm Description

For ease of presentation, we define several significant terminologies below. The tags on target and non-target objects are called *target* and *non-target tags*, respectively. A time slot involving only one object's information is referred to as a *homogeneous slot*, while those involving w ($w \geq 2$) objects' information are called *w-heterogeneous slots*. For instance, in Fig. 2, slot 13 is homogeneous, whereas slot 10 is a 3-heterogeneous slot. A slot t_1 is defined as a *child slot* of another slot t_2 when the information of the object transmitted in t_2 can be parsed by

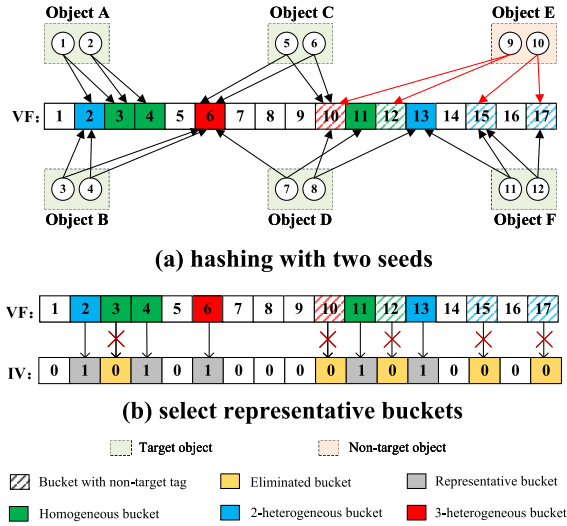


Fig. 3. Representative bucket selection at the reader side.

performing collision peeling with the information transmitted in t_1 . Conversely, t_2 is a *parent slot* of t_1 . For instance, in Fig. 2, slot 2 is a parent slot of slot 1. Notably, both homogeneous and heterogeneous slots can be chosen as representative slots.

BCP consists of multiple rounds, each having three steps:

Step 1. Selecting representative buckets: This step is to select a representative bucket for each target object. The reader constructs a virtual frame (VF) and uses it to devise an indication vector (IV) to indicate the representative slots selected in this round. Notably, the VF is just a bit vector rather than an executed frame. Slots in both the VF and IV are called *buckets*. Specifically, as illustrated in Fig. 3(a), the reader exploits the uniform hashing function $H[(ID, s_i) \bmod f]$ and a set of $k = 2$ seeds $\{s_i\}_{i=1}^2$ to hash each tag ID to 2 repeatable buckets within an VF of length 17. Objects A, B, C, D, and F are target objects, while object E is non-target.

Subsequently, we classify all buckets in the VF into two categories: *representative buckets* and *eliminated buckets*. Initially, any bucket that is mapped by non-target tags is marked as eliminated. In Fig. 3, buckets 10, 12, 15, and 17 are marked eliminated. Next, for each target object that has at least one unmarked homogeneous bucket, we randomly choose one of these buckets to serve as its representative bucket. In Fig. 3, unmarked homogeneous buckets 4 and 11 are marked as the representative buckets for objects A and D, respectively. For the remaining target objects that have not yet been assigned representative buckets, we consider unmarked buckets that can be parsed through collision peeling with already marked representative buckets. If such buckets exist, one is randomly selected as the representative bucket for the corresponding object. For instance, in Fig. 3, unmarked bucket 2 is marked as the representative bucket of object B, as object B's information can be extracted from this bucket via collision peeling with bucket 4. Similarly, unmarked buckets 6 and 13 are marked as representative buckets for objects C and F, respectively. We obtain an IV by setting the values of representative buckets to "1" and all others to "0", as shown in Fig. 3(b).

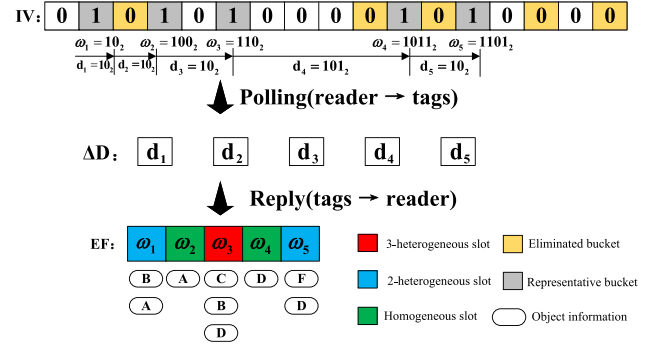


Fig. 4. Target object information response from target tags.

Step 2. Incremental polling and manufacturing of an executed frame (EF): This step begins by sorting the representative buckets (i.e., those with a value of "1" in the constructed IV) in ascending order based on their indexes in the IV. Let $\Omega = \{\omega_1, \omega_2, \omega_3, \dots, \omega_m\}$ denote the index sequence of the sorted buckets. For $1 \leq i \leq m$, we define $\Delta d_i = \omega_i - \omega_{i-1}$, then obtain an incremental sequence as $\Delta D = \{\Delta d_1, \Delta d_2, \Delta d_3, \dots, \Delta d_m\}$, where $\Delta d_1 = \omega_1$.

The reader broadcasts the commands and parameters, including $\{s_i\}_{i=1}^k$, f , and $H(\cdot)$, involved in Step 1, to all tags in the system. Upon receiving the information, each tag computes k repeatable bucket indexes in the IV according to Step 1. For instance, in Fig. 3, tag 1 gets bucket indexes 2 and 3, and tag 5 obtains indexes 6 and 10. Each tag then maintains a recorder R , initialized to 0. The reader sequentially broadcasts each Δd_i in ΔD to all tags, one slot at a time. Upon receiving Δd_i , each tag updates its recorder by setting $R \leftarrow R + \Delta d_i$, and checks if any of its indexes match R . If there is a match, the tag responds to the reader in the current slot; otherwise, it waits for the next update in the subsequent slot. In Fig. 4, the reader broadcasts $\Delta d_1 = 2$ to all tags in the first slot. They then update their recorders to 2. Only tags 1, 3, and 4, which have index 2, respond with the object information (i.e., objects A and B) they carry in this slot. Similarly, in the second slot, after broadcasting $\Delta d_2 = 2$, the reader receives object A's information from tag 2. By fully broadcasting ΔD , the reader obtains an EF comprising 5 slots, each corresponding to a unique representative bucket, as shown in Fig. 4. Hence, tags mapped to representative buckets respond in the EF, while those mapped to eliminated buckets remain silent.

Step 3. Collecting object information: In this step, the reader extracts target objects' l -bit information from the EF using collision peeling. As shown in Fig. 5, the reader directly obtains the l -bit information of objects A and D from the 2nd and 4th slots in the EF, as both slots are homogeneous. For object B, the reader performs collision peeling with the 2nd slot to gather its information from the 1st slot, since the 1st slot serves as a parent slot of the 2nd slot. Similarly, object F's information is recovered from the 5th slot via collision peeling with the 4th slot. The reader also gathers object C's information from the 3rd slot, using collision peeling with both the 1st and 4th slots. Through this process, the l -bit information of the five target objects is successfully collected.

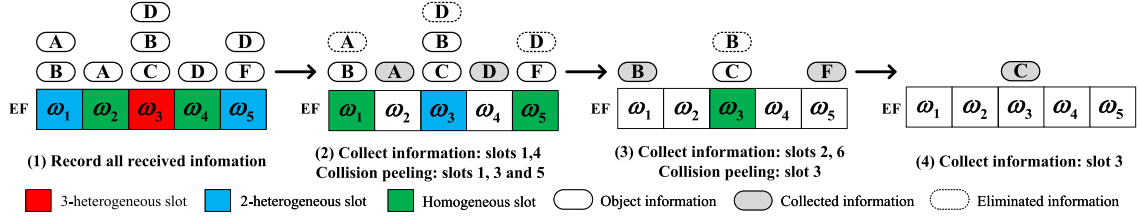


Fig. 5. Target object information collection based on collision peeling at the reader side.

Target objects in $\mathcal{T}$ whose l -bit information remains uncollected in this round are carried over to the next round. The l -bit information of all target objects in $\mathcal{T}$ is collected by iteratively repeating the above procedure round by round.

C. Performance Analysis

The execution time of an RFID information collection algorithm can be divided into two main components: the tag-to-reader uplink and reader-to-tag downlink communications. The uplink communication involves only the transmission of object information, which depends only on the length of the object-level information. In contrast, the downlink communication includes commands broadcast by the reader in each round, which can vary significantly depending on the algorithm's execution mode. To estimate the execution time of BCP, we first consider the number of execution rounds.

Consider any object X and any round $\mathcal{R}$ in the BCP algorithm. Let E represent the total number of objects participating in the round, r the ratio of target objects to all objects, e the number of tags on each object, f the length of the constructed VF, and k the number of hash seeds employed. The probability that any tag on X maps to a specific slot is k/f . For any given object, the probability that at least one of the tags attached to it maps to any slot is

$$p = 1 - \left(1 - \frac{k}{f}\right)^e. \quad (1)$$

In round $\mathcal{R}$, during the construction of the IV, we assume that tags are assigned to buckets in the following order: homogeneous buckets, 2-heterogeneous buckets, 3-heterogeneous buckets, etc. Notably, homogeneous buckets can be treated as 1-heterogeneous buckets. Let P_i represent the probability of any i -heterogeneous bucket being selected as the representative bucket, where i is the degree of heterogeneity. For i -heterogeneous buckets, after excluding all heterogeneous buckets up to $i-1$ (along with the tags assigned to them and the associated objects), the process of marking i -heterogeneous buckets can be approximated as marking homogeneous (1-heterogeneous) buckets. Thus,

$$P_i = r_i E_i \cdot p(1-p)^{E_i-1}, \quad (2)$$

where $r_i \triangleq \frac{rE - (E - E_i)}{E_i}$ and $E_i \triangleq E_{i-1} - fP_{i-1}$ for all $i \geq 2$. The probability that an object in round $\mathcal{R}$ of BCP algorithm has

a representative bucket is given by

$$P = \frac{f \sum P_i}{rE} = \frac{f \sum [r_i E_i \cdot p(1-p)^{E_i-1}]}{rE}. \quad (3)$$

Therefore, the expected number of rounds that object X will participate in is

$$\begin{aligned} h &= 1 \times P + 2 \times (1-P)P + \dots + n \times (1-P)^{n-1}P + \dots \\ &= \sum_{i=0}^{\infty} P(1-P)^i + \sum_{i=1}^{\infty} P(1-P)^i + \sum_{i=2}^{\infty} P(1-P)^i + \dots \\ &= 1 + (1-P) + (1-P)^2 + \dots + (1-P)^n + \dots \\ &= \frac{1}{P} = \frac{rE}{f \sum [r_i E_i \cdot p(1-p)^{E_i-1}]} \end{aligned}$$

Next, we will calculate the average length Δd_{avg} of the polling vectors during the incremental polling phase. Notably, the binary representation of Δd requires $\lceil \log_2(\Delta d) \rceil$ bits. To analyze the expectation of Δd_{avg} , we assume that the distribution of the representative buckets in IV follows a geometric distribution $\text{Geom}(P)$.

Since Δd depends on the interval between two adjacent representative buckets, the probability distribution of Δd is given by $\Pr(\Delta d = x) = (1-P)^{x-1}P$, where P is determined by (3). Hence, the expectation of Δd_{avg} is calculated as

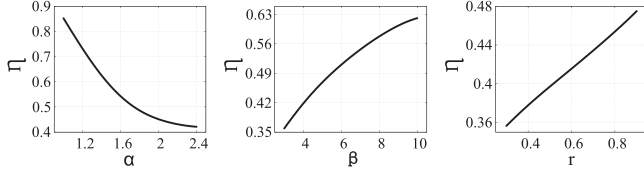
$$E[\Delta d_{\text{avg}}] = P \sum_{x=1}^{\infty} \lceil \log_2(x) \rceil (1-P)^{x-1}. \quad (4)$$

Algebraically, we obtain $E[\Delta d_{\text{avg}}] \approx \frac{-\log_2(P)}{1-P}$. To facilitate subsequent analysis, we define the following parameters:

- Time required for the reader to transmit 1 b: $t_{r \rightarrow t}$
- Time required for each tag to transmit 1 b: $t_{t \rightarrow r}$
- Average length of each command transmitted by the reader in each execution round: L_{avg}
- Number of target objects: T ($T = rE$)

With these definitions, the expected execution time of BCP, denoted by T_{BCP} , is expressed by

$$\begin{aligned} T_{\text{BCP}} &= (T \Delta d_{\text{avg}} + h L_{\text{avg}}) t_{r \rightarrow t} + T l t_{t \rightarrow r} \\ &= \left\{ TP \sum_{x=1}^{\infty} \lceil \log_2(x) \rceil (1-P)^{x-1} \right. \\ &\quad \left. + \frac{L_{\text{avg}} T}{f \sum (r_i E_i \cdot p(1-p)^{E_i-1})} \right\} t_{r \rightarrow t} + T l t_{t \rightarrow r} \end{aligned}$$

Fig. 6. Variation trend of η w.r.t various parameters.

$$\approx T \left\{ t_{r \rightarrow t} \left[\frac{-\log_2(P)}{1-P} + \frac{L_{avg}}{f \sum (r_i E_i \cdot p(1-p)^{E_i-1})} \right] + t_{t \rightarrow r} \right\}.$$

Notably, T_{BCP} depends mainly on P . To further estimate the value of P , we define two important ratios: $\alpha \triangleq f/E$ and $\beta \triangleq pE$. Hence,

$$P_1 = rE \cdot p(1-p)^{E-1} = r\beta(1-p)^{\frac{\beta}{p}-1} \approx r\beta e^{-\beta}. \quad (5)$$

Furthermore, $\beta_i \triangleq pE_i$, $P_i \approx r_i \beta_i e^{-\beta_i}$, and

$$P = \frac{f \sum P_i}{rE} \approx \frac{\alpha \sum r_i \beta_i e^{-\beta_i}}{r}. \quad (6)$$

We compare BCP with the best state-of-the-art algorithm applicable to our problem, called Two-Phase Sampling Protocol (TPS). TPS performs information collection in a round-by-round manner, each round only consisting of singleton slots constructed by a hashing function, achieving a maximum collection efficiency of e^{-1} per round. The collection efficiency P in each round for BCP, as shown in (6), is dominated by r , α and β . Let T_{BCP} be the expected execution time of BCP. To assess the performance improvement of BCP over TPS, we define the following metric:

$$\eta = 1 - \frac{T_{BCP}}{T_{TPS}}. \quad (7)$$

Fig. 6 demonstrates the trend of η under different parameters. We observe that BCP consistently outperforms TPS in all circumstances, with a minimum performance improvement of approximately 35%.

IV. TARGET OBJECT INFORMATION COLLECTION BASED ON ENHANCED BCP (E-BCP)

In this section, we introduce another information collection algorithm, Enhanced BCP (E-BCP), to further enhance system throughput and reduce aggregate time overhead.

A. Algorithm Description

By examining the BCP algorithm, we find that increasing the proportion of representative buckets within each VF can notably reduce the number of information collection rounds, diminishing the aggregate time overhead. The selection of each representative bucket in a round is based on those already marked as representative buckets. Any unmarked buckets are eliminated from consideration in that round. Some of them, however, can still be leveraged to convey target object information through

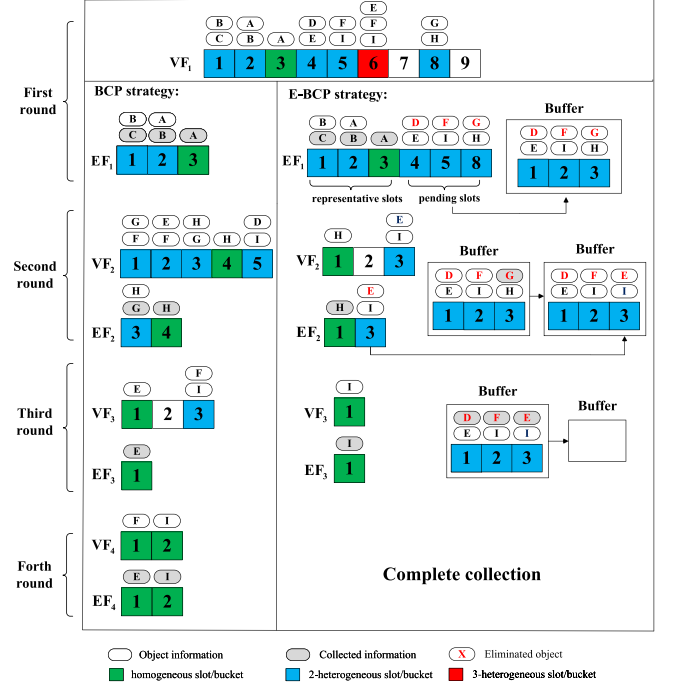


Fig. 7. Comparison between BCP and E-BCP.

collision peeling with representative buckets selected in subsequent rounds. For example, as depicted in Fig. 7, the BCP algorithm constructs a VF_1 in the first round, where bucket 5 will not be selected as a representative bucket as it cannot be parsed using collision peeling in this round. Nevertheless, once we get the information of either of the target objects F and I in bucket 5 in subsequent rounds, we can then acquire the information of the other target object in this bucket. Hence, a pivotal optimization challenge in the BCP algorithm is to select as many representative buckets as possible in each round while ensuring they can all be successfully parsed.

To tackle this challenge, we introduce the E-BCP algorithm, which builds upon the BCP algorithm's framework but leverages a buffer management mechanism to optimize the selection of representative buckets within each VF. Specifically, target objects (e.g., object D mapped to bucket 4 in VF_1) whose information cannot be currently parsed through collision peeling are called *unknown objects*. Buckets (e.g., the bucket 4 in VF_1), which contain information of at least two unknown target objects, are termed *unknown buckets*. We introduce a category of buckets, called *pending buckets*, defined as follows: In each VF, an unknown bucket is classified as a pending bucket if it contains the information of at least one unknown target object not covered by any other unknown bucket. For example, in Fig. 7, all objects are target objects, among which objects D, E, F, I, G, and H are unknown objects. In VF_1 , buckets 4, 5, and 8 are identified as pending buckets. In contrast, bucket 6 is not considered as a pending bucket, as the information it contains is fully covered by buckets 4 and 5. With these definitions, we use Fig. 7 as an illustrative example to elaborate on the E-BCP algorithm as follows:

- *First round:* The algorithm begins by selecting representative buckets using the same strategy as used in the BCP algorithm, i.e., marking buckets 1, 2, and 3 in VF_1 as representative buckets. It then marks all pending buckets, including buckets 4, 5, and 8, as representative buckets. Subsequently, the algorithm performs an information collection operation identical to that of the BCP algorithm. Since each representative bucket corresponds to a representative slot in the EF, the combined information in each of these pending buckets 4, 5, and 8 is also collected. We refer to those representative slots corresponding to pending buckets as *pending slots*. Given that these pending slots cannot be parsed through collision peeling in the current round, the E-BCP algorithm adds them to a buffer. In the buffer, each pending slot is associated with (i) a unique *eliminated object*, which is randomly selected from the unknown objects mapped to that pending slot and will be excluded from participating in the next round; and (ii) a set of *pending objects*, which includes the remaining unknown objects mapped to the same pending slot whose information is still unknown at the moment this pending slot is buffered. For example, in Fig. 7, objects D, F, and G are selected as eliminated objects in the first round, which are marked in red in EF_1 .
- *Second round:* Analogously, the E-BCP algorithm marks buckets 1 and 3 in VF_2 as representative buckets. As the information of target object H can be extracted in representative slot 1 of EF_2 , the pending slot 3 in the buffer is then parsed through collision peeling with the information of target object H. Then, the pending slot 3 is removed from the buffer. Conversely, as bucket 3 is a pending bucket to which target objects E and I are mapped, the pending slot in EF_2 that it is mapped to is added to the buffer. After that, object E is selected to be excluded from the next round.
- *Third round:* After extracting the information of target object I in representative slot 1 of EF_3 , the algorithm can parse all pending slots in the buffer via collision peeling. Hence, the information of all target objects is successfully collected by repeating the above process iteratively.

It is worth noting that: the algorithm guarantees that each pending slot in the buffer can be successfully parsed, because target objects whose information has been successfully parsed will not participate in subsequent rounds, which means that as more and more target objects have their information parsed, the information in each pending slot will subsequently be parsed using collision peeling. From Fig. 7, it can be observed that the E-BCP algorithm outperforms the BCP algorithm by reducing the number of information collection rounds by one. The performance gap becomes even more pronounced in large-scale scenarios.

B. Performance Analysis

1) *Computational and Memory Overhead At the Reader:* E-BCP temporarily stores those *pending slots* that cannot be parsed through collision peeling in the current round, and resolves them later as more object information is obtained. To keep the buffer workload bounded in large-scale settings, we

enforce a bounded-pending rule: a collision slot is marked as *pending* only if it contains no more than a constant number w of unknown objects. Since each pending slot is indexed by a distinct eliminated object, the total number of buffered pending slots in a task is at most the number of target objects E .

Computational overhead: For each pending slot, the reader stores an l -bit received signal (used for collision peeling) together with a constant-size state that tracks how many of the objects in this slot are still unknown. When an unknown object becomes known in later rounds, the reader updates every buffered pending slot that contains this object by performing one collision-peeling update on its stored l -bit signal and refreshing the associated constant-size state. Each such update scans an l -bit signal and involves only constant-time state updates, so the per-update cost is $O(l)$.

Under the bounded-pending rule, each pending slot contains at most w unknown objects, and thus it can be updated at most $(w - 1)$ times before it becomes parsable. With at most E pending slots buffered in total, the overall reader-side computation is bounded by $O(E \cdot (w - 1) \cdot l) = O(wEl)$.

Memory overhead: In the worst case, the buffer contains at most E pending slots. For each pending slot, the reader stores (i) a constant-size header (including the eliminated-object ID and a counter indicating how many objects in this slot remain unknown) and (ii) one stored l -bit signal for collision peeling. This contributes $O(E) + O(E \cdot l)$ memory in total.

In addition, to enable the above updates, for each pending slot the reader needs to record the IDs of the other objects that are involved in this slot (besides its eliminated object). Under the bounded-pending rule, this number is at most $(w - 1)$ per pending slot, contributing at most $O(wE)$ additional memory. Therefore, the total buffer memory is bounded by $O(E \cdot l) + O(wE) = O(E(l + w))$.

2) *Expected Execution Time:* The buffer management in E-BCP requires the reader to maintain pending-slot states and perform lightweight bookkeeping and peeling updates locally. Throughout this paper, we treat such reader-side processing as negligible by assuming that the reader has sufficient computing capability, while the overall latency in COTS RFID systems is dominated by over-the-air transmissions. Hence, the execution-time analysis in this subsection focuses on the *communication overhead*, i.e., the uplink tag responses and the downlink reader commands.

In a given round, a u -unknown bucket is defined as a bucket containing u unknown target objects. In this subsection, the representative buckets refer only to the buckets that can be parsed in the current round, while those that can be parsed in the subsequent round are referred to as pending buckets. For any round $\mathcal{R}$, let N_u and P_{N_u} denote the number of u -unknown buckets and the probability of a u -unknown bucket occurring in that round, respectively. Hence, the expectation of N_u is given by $\mathbb{E}[N_u] = fP_{N_u}$. we have

$$P_{N_u} = (r_N)^u \binom{E - rEP_f}{u} p^u (1 - p)^{E - rEP_f - u},$$

where P_f represents the probability that a bucket is marked as a representative bucket in round $\mathcal{R}$ (cf. Equation (3)). r_N denotes the ratio of the number of the target objects that have not yet

been assigned representative buckets after round $\mathcal{R}$ to E , which is mathematically defined as $r_N \triangleq rE(1 - P_f)/E$.

Let S_1 and S_2 denote the sets of eliminated and pending objects in round $\mathcal{R}$, respectively. We use the tuple $St = (|S_1| = p, |S_2| = q)$ to represent the state where $|S_1| = p$ and $|S_2| = q$. In round $\mathcal{R}$ with $St = (|S_1| = p, |S_2| = q)$, for any unknown bucket, if any of its unknown target objects are in either S_1 or S_2 , it is marked as an eliminated bucket, and St remains unchanged. Otherwise, St will transition from (p, q) to $(p + 1, q + i)$, where i is the number of new pending objects generated in that bucket. Let Pe_u be the probability that a u -unknown bucket is eliminated in round $\mathcal{R}$, and let $P_u\{St : St_1 \rightarrow St_2\}$ denote the probability that St converts from state St_1 to state St_2 in round $\mathcal{R}$. We have

$$\begin{aligned} Pe_u &= P_u\{St : (|S_1| = p, |S_2| = q) \rightarrow (|S_1| = p, |S_2| = q)\} \\ &= \left(\frac{q}{N}\right)^u + \sum_{i=1}^u \left[\binom{u}{i} \left(\frac{p}{N}\right)^i \left(1 - \frac{p}{N}\right)^{u-i} \right] \\ &= 1 - \left(1 - \frac{p}{N}\right)^u + \left(\frac{q}{N}\right)^u, \end{aligned} \quad (8)$$

where $xN \triangleq rE(1 - P_f)$. We further obtain

$$\begin{aligned} P_u\{St : (|S_1| = p, |S_2| = q) \rightarrow (|S_1| = p + 1, |S_2| = q + i)\} \\ = \binom{N}{i + 1} \left(1 - \frac{p + q}{N}\right)^{i+1} \left(\frac{q}{N}\right)^{N-i-1}. \end{aligned} \quad (9)$$

Next, we construct a Markov chain $\mathcal{X}$ to model the bucket-by-bucket evolution of the state $St = (|S_1|, |S_2|)$ within round $\mathcal{R}$. This is because, under the transition probabilities Pe_u and $P_u\{St_1 \rightarrow St_2\}$ derived above, the effect of scanning the next u -unknown bucket depends only on the current $(|S_1|, |S_2|)$, rather than on the specific identities of objects, which allows us to aggregate the randomness over the $\mathbb{E}[N_2]$ buckets without enumerating all selection combinations. It is worth noting that this Markov-chain calculation is used only for performance analysis and is not computed during the online execution of E-BCP.

Given the initial state of St as $St_0 = (|S_1| = 0, |S_2| = 0)$, for any state $(|S_1|, |S_2|)$, we have $|S_1| > 0$, $1 \leq |S_2| \leq |S_1|$. Therefore, there are a total of $(1 + \mathbb{E}[N_2])\mathbb{E}[N_2]/2 + 1$ states in $\mathcal{X}$. Let $\mathcal{M}$ denote the transition matrix of $\mathcal{X}$. We denote by $\mathcal{M}^{\mathbb{E}[N_2]}(1, i)$ the 1st row and i -th column of $\mathcal{M}^{\mathbb{E}[N_2]}$, which corresponding to the transition state $(|S_1| = p_i, |S_2| = q_i)$. The expectations of $|S_1|$ and $|S_2|$ after processing $\mathbb{E}[N_2]$ 2-unknown buckets are then given by:

$$\mathbb{E}[|S_1|_2] = \sum_{i=1}^S \left(\mathcal{M}^{\mathbb{E}[N_2]}(1, i) \times p_i \right), \quad (10)$$

$$\mathbb{E}[|S_2|_2] = \sum_{i=1}^S \left(\mathcal{M}^{\mathbb{E}[N_2]}(1, i) \times q_i \right). \quad (11)$$

where $|S_1|_i$ indicates the number of pending buckets after checking the i -unknown buckets.

Starting from $(\mathbb{E}[|S_1|_2], \mathbb{E}[|S_2|_2])$, we proceed similarly for $u = 3, 4, \dots, w$ (where w is the maximum number of unknown objects mapped to the same bucket in round $\mathcal{R}$) and obtain the

final expected state $(\mathbb{E}[|S_1|_w], \mathbb{E}[|S_2|_w])$. Hence, the probability of a pending bucket is $P_t = \mathbb{E}[|S_1|]/f = \mathbb{E}[|S_1|_w]/f$.

In summary, in any round, the probability that any target object is eliminated from subsequent rounds is expressed as

$$P = P_f + P_t. \quad (12)$$

Therefore, under our E-BCP algorithm, the expected number of rounds in which a target object participates is $h = 1/P = 1/P_f + P_t$.

Hence, the expected execution time of E-BCP is given by:

$$\begin{aligned} T_{E-BCP} &= (T\Delta d_{avg} + hL_{avg})t_{r \rightarrow t} + Tlt_{t \rightarrow r} \\ &= \left\{ T(P_f + P_t) \sum_{x=1}^{\infty} [\log_2(x)] [1 - (P_f + P_t)]^{x-1} \right. \\ &\quad \left. + \frac{L_{avg}}{P_f + P_t} \right\} t_{r \rightarrow t} + Tlt_{t \rightarrow r}. \end{aligned} \quad (13)$$

3) Performance Comparison: To compare the performance between our E-BCP and BCP algorithms, we consider an overhead-maximizing scenario for E-BCP: In any given round $\mathcal{R}$, we are restricted to selecting pending buckets solely from the maximum of $\mathbb{E}[N_2]$ 2-unknown buckets generated in that round, which results in the minimum number of pending buckets in that round. The expected number of target objects mapped to the $\mathbb{E}[N_2]$ buckets is $N[1 - (1 - \frac{1}{N})^{2\mathbb{E}[N_2]}]$. Thus, the probability of selecting a pending bucket among all buckets involved in round $\mathcal{R}$ is at least

$$P_t^- = \frac{N \left[1 - \left(1 - \frac{1}{N}\right)^{2\mathbb{E}[N_2]} \right]}{2f}. \quad (14)$$

Let m be the maximum number of tags mapped to the same bucket during a single mapping under this scenario. We have

$$\begin{aligned} P_{N_2} &= (r_N)^2 \left(\frac{E - rE(P_f + P_s)}{2} \right) p^2 (1 - p)^{E - rE(P_f + P_s) - 2} \\ &\approx \frac{r_m^2 \beta_m^2 e^{-\beta_m}}{2}, \end{aligned} \quad (15)$$

where $\beta_m = pE_m = p[E - (1 - P_m)f] = pN$, f is the frame size, and P_i is the probability of any i -heterogeneous bucket being selected as the representative bucket. Additionally, $E_i = E_{i-1} - fP_i$. Consequently, $P_t^- \approx \frac{\beta_m(1 - e^{-\alpha\beta_m r_m^2 e^{-\beta_m}})}{2\alpha}$.

To assess the performance improvement of E-BCP over BCP, we introduce the following parameter:

$$\lambda = 1 - \frac{T_{E-BCP}}{T_{BCP}}. \quad (16)$$

Combing the above analysis, the variation of λ with respect to the parameters r , α , and β is shown in Fig. 8. It is evident that, even when considering only one type of pending slots, E-BCP still achieves a performance improvement of up to 35% compared to BCP.

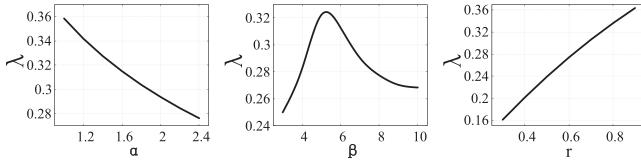


Fig. 8. Variation trend of λ w.r.t various parameters.

V. DISCUSSION

A. C1G2-Compatible Implementation of BCP/E-BCP

We discuss how to make our algorithms compatible with the RFID C1G2 [27] protocol. As a standard communication mechanism [19], the C1G2 protocol specifies a cycle of the reader-tag dialog including three phases: SELECT, INVENTORY, and ACCESS. SELECT selects a subset of tags to participate in the subsequent phases. INVENTORY then initializes multiple frames to identify the selected tags. ACCESS further performs transactions with the identified tags, e.g., writing and reading. Our algorithms BCP/E-BCP can be implemented within the C1G2 protocol through the following three steps.

Step 1. Selecting a representative bucket for each target object: The SELECT phase issues a command, abstractly defined as $\text{SELECT}\{\text{POINTER}, \text{LENGTH}, \text{MASK}\}$, where POINTER and LENGTH specify the starting bit and the number of consecutive bits in the tag memory to compare with a mask, and MASK defines the mask. Tags whose memory contents match the specified MASK are selected to participate in the subsequent INVENTORY and ACCESS phases. In BCP/E-BCP, the reader first performs a hashing operation that maps all tags to an IV. For any tag g_t , let $V(g_t)$ denote its hash value, which serves as the bucket index of g_t in the IV. By the design of BCP/E-BCP, tags whose hash values belong to the sequence $\Omega = \{\omega_1, \omega_2, \omega_3, \dots, \omega_m\}$ are target tags, each mapped to a representative bucket.

Step 2. Incremental polling target tags: In the INVENTORY phase, all representative buckets identified in Step 1 are integrated into an EF. For any target tag g_a mapped to a representative bucket in the EF, BCP/E-BCP preloads its hash value $V(g_a)$ into a designated memory region of tag g_a , starting at bit position B_s with a length of $\lceil \log_2(V(g_a) + 1) \rceil$ bits. Accordingly, BCP/E-BCP issues a polling vector $\text{SELECT}\{B_s, \lceil \log_2(V(g_a) + 1) \rceil, V(g_a)\}$ to select tag g_a in the corresponding slot. During the ACCESS phase, the reader receives signals transmitted by target tags on a slot-by-slot basis.

Step 3. Decoding object information: The reader extracts the target object information from the received signals using collision peeling.

B. Handling Unknown Tags and Unreliable Channels

This subsection discusses how BCP/E-BCP can be adapted to handle unknown tags and unreliable channels.

To handle unknown tags, in each information collection round, BCP/E-BCP first performs an efficient ID collection procedure over all tags. The collected IDs are then used to gather

the l -bit information from the target tags, while excluding tags whose IDs have not yet been collected in this round. Target tags whose information has already been gathered do not participate in subsequent rounds. This process is repeated iteratively, round by round, until the information of all target tags has been fully collected.

To handle unreliable communication channels, BCP/E-BCP leverages the C1G2 communication protocol to identify slots affected by non-ideal conditions. Specifically, the protocol defines commands transmitted between the reader and each tag with stringent timing and intervals. If the reader receives a backscattered signal from a tag that is delayed, early, or erroneous, it can infer that the corresponding slot may be affected by an imperfect channel. In such cases, the reader can initialize an additional slot to collect information from this tag. Similarly, if channel imperfections cause larger synchronization offsets among the backscattered signals of different tags, the reader can detect them by examining the timing or duration of the received signals. Additionally, synchronization issues can be identified by verifying the length of the received information, as such offsets may result in abnormally shorter or longer combined signals.

VI. EXPERIMENTAL EVALUATION

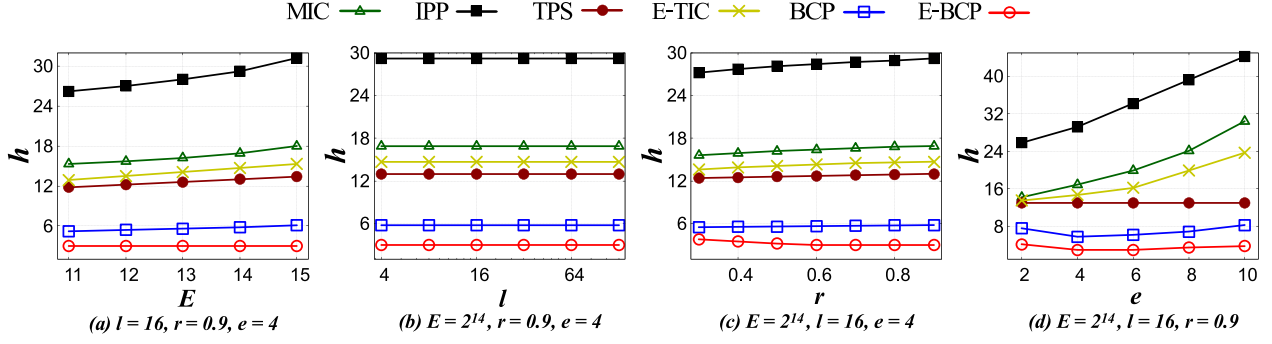
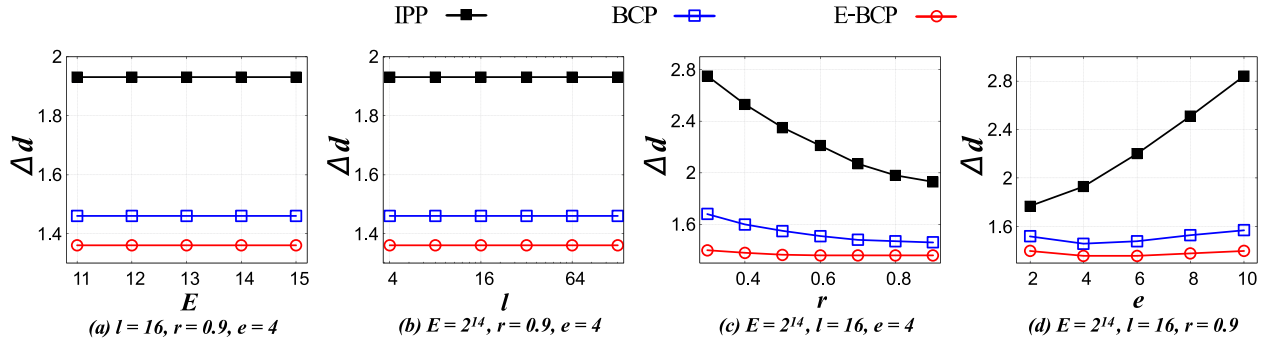
A. Experimental Methodology

Experimental setup: The simulation is configured according to the C1G2 standard specification. Any two consecutive transmissions—whether from the reader to the tag or vice versa—are separated by an interval of $100 \mu\text{s}$. The reader transmits commands at a rate of 26.5 Kb/s , while tags transmit information at 53 Kb/s . Consequently, the time required for the reader to transmit one bit is $37.74 \mu\text{s}$, and for a tag, it is $18.87 \mu\text{s}$. The time required for a tag to transmit a piece of l -bit object information is calculated as the product of l and 18.87 , plus the inter-transmission interval. For example, if $l = 16$, we have $t_{inf} = 16 \times 18.87 + 100 = 401.92 \mu\text{s}$.

Baselines: We select four state-of-the-art algorithms as benchmarks, including Multi-hash Information Collection protocol (MIC) [9], Enhanced Target Tags Information Collection protocol (E-TIC) [28], Two-Phase Sampling protocol (TPS) [20], and Incremental Polling Protocol (IPP) [17].

Performance metrics: We track two important performance metrics: the number h of execution rounds and the total execution time T . Additionally, for algorithms that rely on a polling strategy, we also track the average polling vector length, denoted by Δd_{avg} . Since the length of the polling vector in TPS is related to the length of the hash index used, which is significantly longer than that of other polling strategies, we exclude TPS from the comparison of Δd_{avg} .

Parameter setting: To compare the performance of the algorithms described above, we focus on four key parameters: r , l , E and e . We investigate the impact of each parameter on the performance metrics h , Δd_{avg} , and T , while keeping other parameters fixed. All experimental results are averaged over 20 independent runs. Notably, the unit of Δd_{avg} is a bit, while that of T is a second.

Fig. 9. Performance comparison on h under various parameter settings.Fig. 10. Performance comparison on Δd_{avg} under various parameter settings.

B. Experimental Results

Performance comparison on h : As shown in Fig. 9(a), h in MIC, E-TIC, TPS, IPP and BCP increases with linear growth of E , with BCP exhibiting the least performance variation in h . In contrast, h in E-BCP remains relatively stable across different values of E . Additionally, from Fig. 9(b), we observe that variations in l have no noticeable impact on h . In Fig. 9(c), the occupancy ratio of the target object, denoted as r , influences h in a manner similar to E . However, under a small r , E-BCP may require additional rounds of information collection to handle non-target objects, thereby increasing h . Conversely, h in BCP shows a slight increase as r grows. In Fig. 9(d), the trend of h can be classified into three patterns. As e increases, h in MIC, E-TIC, and IPP continues to grow, whereas TPS remains constant across all values of e . In contrast, BCP and E-BCP exhibit a non-monotonic trend in which h initially decreases and then increases. This non-monotonic behavior arises because having either too many or too few tags per object can lead to an excessive or insufficient number of collision slots, thus diminishing the effectiveness of collision peeling. Overall, BCP reduces h by at least 56% compared to the four state-of-the-art solutions, while E-BCP achieves a further reduction of approximately 48% over BCP.

Performance comparison on Δd_{avg} : From Fig. 10, we observe that increases in both E and l have no impact on Δd_{avg} . In contrast, an increase in r leads to a decrease in Δd_{avg} . This

is because only r influences the proportion of target objects that participate in each polling round, which is consistent with our theoretical performance analysis. Moreover, in Fig. 10(d), the impact of parameter e on Δd_{avg} is similar to its effect on h . As e increases, the probability of generating singleton slots in IPP decreases, resulting in a longer polling vector. In contrast, BCP and E-BCP require an appropriate number of tags to maximize the effectiveness of collision peeling. Statistically, as shown in Figs. 10(a) and (b), Δd_{avg} is approximately 1.93 bits for IPP, 1.46 bits for BCP, and 1.36 bits for E-BCP. These results indicate that BCP reduces the latency in transmitting polling vectors by about 24% compared to IPP, and E-BCP achieves a reduction of roughly 30% over IPP.

Performance comparison on T : As shown in Fig. 11, the execution time T in each algorithm dramatically increases with the growth of E , l , and r , albeit at different rates, while the impact of e on T is similar to its effect on h or Δd_{avg} . Consistent with the trends observed in Figs. 9 and 10, Fig. 11 also shows a significant performance improvement of our proposed algorithms over the four baselines. For instance, under the settings $E = 2^{14}$, $l = 16$, $r = 0.9$ and $e = 4$ BCP outperforms the best existing approach TPS by 40%, and E-BCP achieves a reduction of 55% compared to TPS.

Performance comparison between BCP and E-BCP: As shown in Fig. 12, T in both algorithms is positively correlated with β and r , and negatively correlated with α . Under various parameter settings, the performance improvement of E-BCP

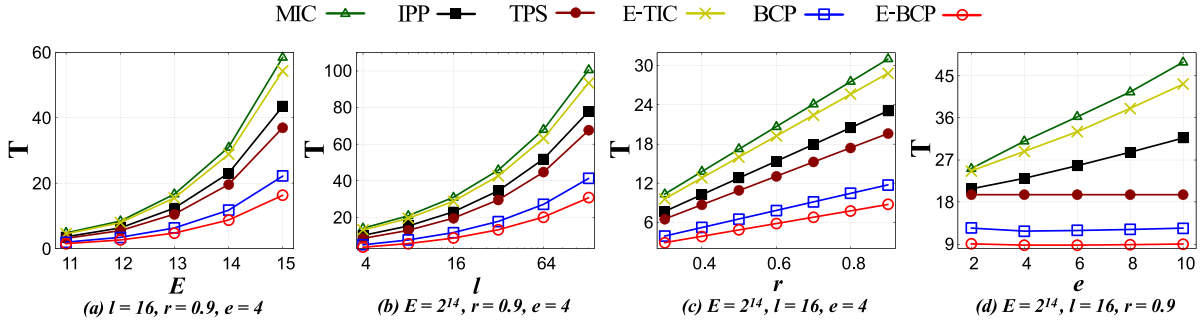


Fig. 11. Performance comparison on T under various parameter settings.

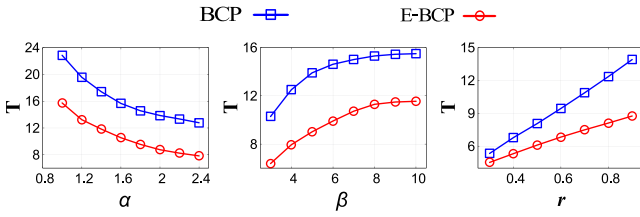


Fig. 12. Performance comparison between BCP and E-BCP.

over BCP slightly exceeds the theoretical values presented in Fig. 8. For instance, when $\alpha = 2$, the theoretical improvement demonstrated in Fig. 8 is 29.36%, whereas the actual observed improvement is roughly 35%. This discrepancy arises because the theoretical analysis in Section IV is a conservative estimate.

Summary: Overall, we can draw the following conclusions from our experimental results: (1) d is independent of E and l , but is significantly influenced by r and e ; (2) T exhibits a positive correlation with both h and d ; (3) the required number h of rounds in our algorithms first decreases and then increases with e , as observed in Fig. 12(d). This indicates that, for a given set of other parameters, there exists an optimal value of e that minimizes h and hence maximizes system throughput; and (4) our algorithms BCP and E-BCP outperform all the simulated state-of-the-art solutions with non-negligible improvement in terms of h , Δd_{avg} , and T . Particularly, E-BCP achieves an average improvement of 50% in T over the best existing algorithm.

VII. RELATED WORK

In RFID systems, information collection has long been recognized as a crucial task. Existing RFID information collection algorithms can generally be classified into two categories.

A. Information Collection Without Unknown Tags

In this scenario, where the reader already knows all tag IDs in advance, the main goal is to collect the information associated with each tag efficiently. Research in this area is typically divided into two settings: single-reader and multi-reader scenarios.

Single-Reader Scenario: [9] proposed the Single-hash Information Collection (SIC) and Multi-hash Information Collection (MIC) algorithms. SIC hashes tag IDs before the reader performs

the FSA process, thereby minimizing the number of transmission bits. MIC extends SIC by using multiple hash functions, which enhances the utilization of available slots. [17] introduced a series of single-polling algorithms, including the Incremental Polling Protocol (IPP), which reduces query overhead by broadcasting the differences between adjacent polling vectors. [18] further developed the Partial Double Polling (PDP) algorithm, which employs a dual-index mechanism to query pairs of tags using a single polling vector, improving efficiency compared to IPP. [20] proposed a tag category information collection protocol called Two-Phase Sampling protocol (TPS), which efficiently samples from a category with minimal overhead by rapidly focusing on a specific category and isolating a tag within it. [26] introduced the near-OPTimal Category-information sampling protocol (OPT-C), which addresses the problem of tag category information collection under missing-tag conditions. By a Bernoulli trial mechanism and a refined sampling protocol, OPT-C achieves execution time close to optimum.

Multi-Reader Scenario: [16] proposed the Bloom Filter-based Information Collection (BIC) algorithm, which uses a bloom filter to filter out tags out of the reader's coverage, improving collection efficiency. [28] designed two efficient algorithms for collecting information on target tags: Target Tags Information Collection (TIC) and Enhanced TIC (E-TIC). TIC operates in two phases: a selection phase, where target tags to query are determined based on test, elimination, and mixed slots, and an information collection phase, where an assignment vector is constructed to guide the information transmission from target tags to the reader. E-TIC builds on TIC by incorporating Multi-Seed Reconciled (MSR) technology. This technology uses multiple seeds to adjust collision slots to singleton slots, thereby improving slot utilization.

B. Information Collection With Unknown Tags

In contrast, when some tag IDs are unknown, the primary challenge is to identify the target tags while filtering out non-target tags accurately. [29] extended the BIC algorithm to enable information collection on target tags with unknown tags. [10] focused on handling unexpected tags by designing the Hierarchical-hashing Data Collection (HDC) and Multi-framed Hierarchical-hashing Data Collection (MHDC) algorithms. HDC uses classification seeds to map target tags to corresponding subsets, eliminating the influence of unexpected

tags. As the time complexity of HDC increases exponentially with the number of tags, MHDC addresses this challenge by dividing large computational tasks into multiple frames, which share computation resources. It constructs shadow vectors to distribute the target tags across different frames and executes the HDC process in each frame. In addition to TPS, [20] also designed the Back-and-Forth Sampling protocol(BFS) for scenarios where tag IDs or category IDs are unknown. By carrying out a step-forward frame and using the step-backward scheme, BFS is able to interrogate only 1.45 tags (close to the lower bound of one tag) on average for each category.

VIII. CONCLUSION

In this paper, we have proposed Backtracking Collision Peeling (BCP), an efficient algorithm for target object information collection in multi-tagged RFID systems, featuring three key innovations. First, BCP selects one slot (allowing even collision slots) for each target object to gather its information. By avoiding high-latency collision elimination, BCP significantly reduces time overhead. Second, BCP leverages the dependencies among the selected slots to recover object information in complex collision slots using data from already resolved slots, offering a novel and effective approach to handling signal collisions. Third, to optimize system throughput, BCP polls tags by broadcasting only incremental changes between polling vectors, rather than the complete vectors. For further improvement, we have introduced an enhanced version E-BCP, which optimizes the utilization of collision slots, reducing the number of required polling rounds. Our experimental results demonstrate that both BCP and E-BCP outperform existing methods by at least 35% in the aggregate execution time, while also exhibiting greater stability and robustness. Regarding further work, we plan to explore other critical yet under-investigated challenges in multi-tagged RFID systems.

REFERENCES

- [1] C. Wang et al., "Spin-antenna: Enhanced 3D motion tracking via spinning antenna based on COTS RFID," *IEEE Trans. Mobile Comput.*, vol. 23, no. 2, pp. 1347–1365, Feb. 2024.
- [2] J. Yin, H. Mao, R. Zhu, and S. Xu, "Parallel missing tag identification for anonymous multiple users RFID systems," in *Proc. 21st Annu. IEEE Int. Conf. Sens., Commun., Netw.*, 2024, pp. 1–9.
- [3] X. Liu, J. Yin, S. Zhang, B. Ding, S. Guo, and K. Wang, "Range-based localization for sparse 3-D sensor networks," *IEEE Internet Things J.*, vol. 6, no. 1, pp. 753–764, Feb. 2019.
- [4] K. Liu, L. Chen, J. Huang, S. Liu, and J. Yu, "Revisiting RFID missing tag identification," in *Proc. IEEE IEEE Conf. Comput. Commun.*, 2022, pp. 710–719.
- [5] J. Yu, W. Gong, J. Liu, L. Chen, K. Wang, and R. Zhang, "Missing tag identification in COTS RFID systems: Bridging the gap between theory and practice," *IEEE Trans. Mobile Comput.*, vol. 19, no. 1, pp. 130–141, Jan. 2020.
- [6] J. Yin, X. Xie, H. Mao, and S. Guo, "Efficient missing key tag identification in large-scale RFID systems: An iterative verification and selection method," *IEEE Trans. Mobile Comput.*, vol. 24, no. 3, pp. 2253–2269, Mar. 2025.
- [7] N. Fescioglu-Unver, S. H. Choi, D. Sheen, and S. Kumara, "RFID in production and service systems: Technology, applications and issues," *Inf. Syst. Front.*, vol. 17, pp. 1369–1380, 2015.
- [8] L. Xie, Y. Yin, A. V. Vasilakos, and S. Lu, "Managing RFID data: Challenges, opportunities and solutions," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 3, pp. 1294–1311, Third Quarter 2014.
- [9] S. Chen, M. Zhang, and B. Xiao, "Efficient information collection protocols for sensor-augmented RFID networks," in *Proc. IEEE INFOCOM*, 2011, pp. 3101–3109.
- [10] X. Liu et al., "Fast RFID sensory data collection: Trade-off between computation and communication costs," *IEEE/ACM Trans. Netw.*, vol. 27, no. 3, pp. 1179–1191, Jun. 2019.
- [11] K. Liu, J. Yu, and L. Chen, "On information collection in multi-tagged COTS RFID systems," *IEEE Trans. Mobile Comput.*, vol. 24, no. 11, pp. 12505–12516, Nov. 2025.
- [12] M. Golsorkhtabamiri and N. Issazadehkojidi, "A distance based RFID reader collision avoidance protocol for dense reader environments," *Wireless Pers. Commun.*, vol. 95, no. 2, pp. 1781–1798, 2017.
- [13] H. Liu, R. Zhang, L. Chen, J. Yu, J. Liu, and J. An, "On fast and reliable missing event detection protocol for multitagged RFID systems," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 10324–10335, Oct. 2020.
- [14] IDTechEx, "RFID progress at Wal-mart," 2005. [Online]. Available: www.idtechex.com/products/en/articles/00000161.asp
- [15] H. Liu et al., "Computation–communication tradeoffs for missing multitagged item detection in RFID networks," *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1252–1264, Jan. 2022.
- [16] H. Yue, C. Zhang, M. Pan, Y. Fang, and S. Chen, "A time-efficient information collection protocol for large-scale RFID systems," in *Proc. 2012 Proc. IEEE INFOCOM*, 2012, pp. 2158–2166.
- [17] J. Liu, B. Xiao, X. Liu, K. Bu, L. Chen, and C. Nie, "Efficient polling-based information collection in RFID systems," *IEEE/ACM Trans. Netw.*, vol. 27, no. 3, pp. 948–961, Jun. 2019.
- [18] K. Lin, H. Chen, N. Yan, Z. Ni, Z. Wang, and J. Yu, "Double polling-based tag information collection for sensor-augmented RFID systems," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 3496–3509, May 2024.
- [19] K. Liu, L. Chen, J. Yu, and H. Cui, "On batch writing in COTS RFID systems," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 3846–3857, May 2024.
- [20] J. Liu, S. Chen, Q. Xiao, M. Chen, B. Xiao, and L. Chen, "Efficient information sampling in multi-category RFID systems," *IEEE/ACM Trans. Netw.*, vol. 27, no. 1, pp. 159–172, Feb. 2019.
- [21] J. Yu, W. Gong, J. Liu, L. Chen, and K. Wang, "On efficient tree-based tag search in large-scale RFID systems," *IEEE/ACM Trans. Netw.*, vol. 27, no. 1, pp. 42–55, Feb. 2019.
- [22] K. Liu, L. Chen, J. Yu, and Z. Jia, "Revisiting RFID missing tag identification: Theoretical foundation and algorithm design," *IEEE/ACM Trans. Netw.*, vol. 32, no. 5, pp. 4056–4066, Oct. 2024.
- [23] X. Xie et al., "Implementation of differential tag sampling for COTS RFID systems," *IEEE Trans. Mobile Comput.*, vol. 19, no. 8, pp. 1848–1861, Aug. 2020.
- [24] J. Liu, X. Chen, X. Liu, X. Zhang, X. Wang, and L. Chen, "On improving write throughput in commodity RFID systems," in *Proc. IEEE Conf. Comput. Commun.*, 2019, pp. 1522–1530.
- [25] J. Su, Z. Sheng, C. Huang, G. Li, A. X. Liu, and Z. Fu, "Identifying RFID tags in collisions," *IEEE/ACM Trans. Netw.*, vol. 31, no. 4, pp. 1507–1520, Aug. 2023.
- [26] X. Wang et al., "A near-optimal category information sampling in RFID systems," *IEEE Trans. Mobile Comput.*, vol. 24, no. 9, pp. 8228–8244, Sep. 2025.
- [27] EPCglobal Inc., *EPC Radio-Frequency Identity Protocols Class-1 Generation-2 UHF RFID Protocol for Communications at 860 MHz–960 MHz*, Version 1.0.9, Jan. 26, 2005.
- [28] X. Liu, J. Yin, S. Zhang, B. Xiao, and B. Ou, "Time-efficient target tags information collection in large-scale RFID systems," *IEEE Trans. Mobile Comput.*, vol. 20, no. 9, pp. 2891–2905, Sep. 2021.
- [29] H. Yue, C. Zhang, M. Pan, Y. Fang, and S. Chen, "Unknown-target information collection in sensor-enabled RFID systems," *IEEE/ACM Trans. Netw.*, vol. 22, no. 4, pp. 1164–1175, Aug. 2014.



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